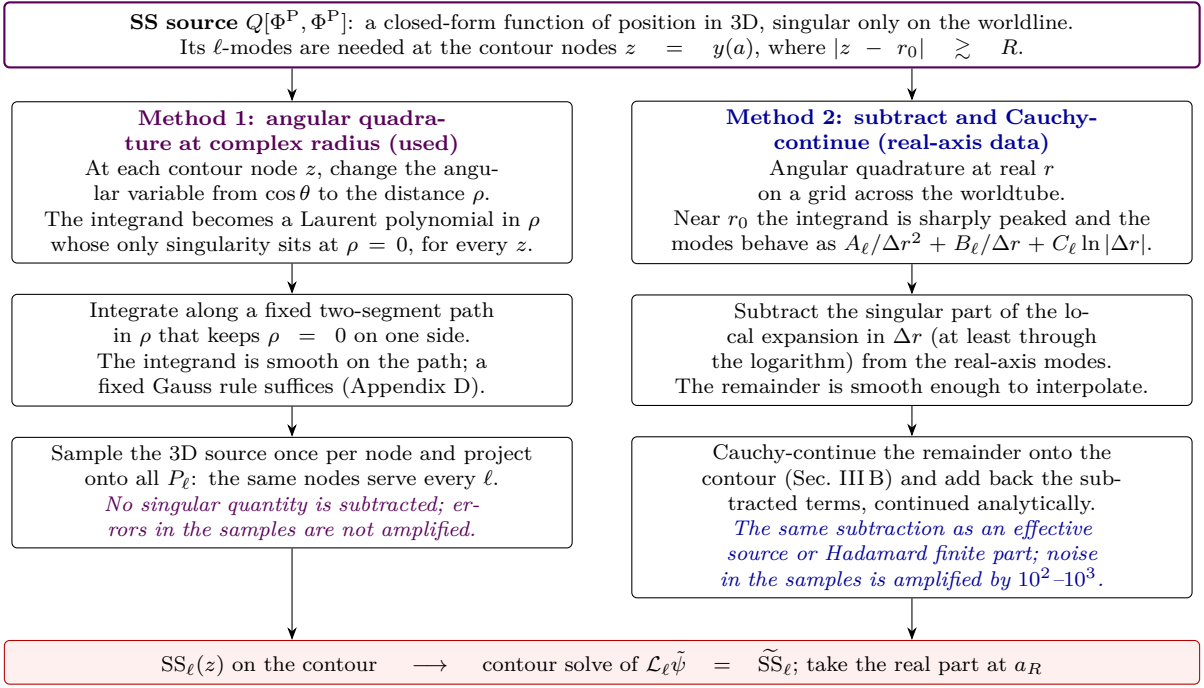

SS source $Q[\Phi^{\mathrm{P}}, \Phi^{\mathrm{P}}]$: a closed-form function of position in 3D, singular only on the worldline.
Its ℓ-modes are needed at the contour nodes $z = y(a)$, where $|z - r_0| \gtrsim R$.
Method 1: angular quadrature at complex radius (used)
At each contour node z, change the angular variable from $\cos\theta$ to the distance ρ.
The integrand becomes a Laurent polynomial in ρ whose only singularity sits at $\rho = 0$, for every z.
Integrate along a fixed two-segment path in ρ that keeps $\rho = 0$ on one side.
The integrand is smooth on the path; a fixed Gauss rule suffices (Appendix D).
Sample the 3D source once per node and project onto all P_ℓ: the same nodes serve every ℓ.
No singular quantity is subtracted; errors in the samples are not amplified.
Method 2: subtract and Cauchy-continue (real-axis data)
Angular quadrature at real r on a grid across the worldtube.
Near r_0 the integrand is sharply peaked and the modes behave as $A_\ell/\Delta r^2 + B_\ell/\Delta r + C_\ell \ln|\Delta r|$.
Subtract the singular part of the local expansion in Δr (at least through the logarithm) from the real-axis modes.
The remainder is smooth enough to interpolate.
Cauchy-continue the remainder onto the contour (Sec. III B) and add back the subtracted terms, continued analytically.
The same subtraction as an effective source or Hadamard finite part; noise in the samples is amplified by 10^2–10^3.
$\mathrm{SS}_\ell(z)$ on the contour $\longrightarrow$ contour solve of $\mathcal{L}_\ell\tilde{\psi} = \widetilde{\mathrm{SS}}_\ell$; take the real part at a_R